EEE103 ELECTRICAL CIRCUITS

WEEK4-HANDY CIRCUIT ANALYSIS TECHNIQUES

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CONTENT

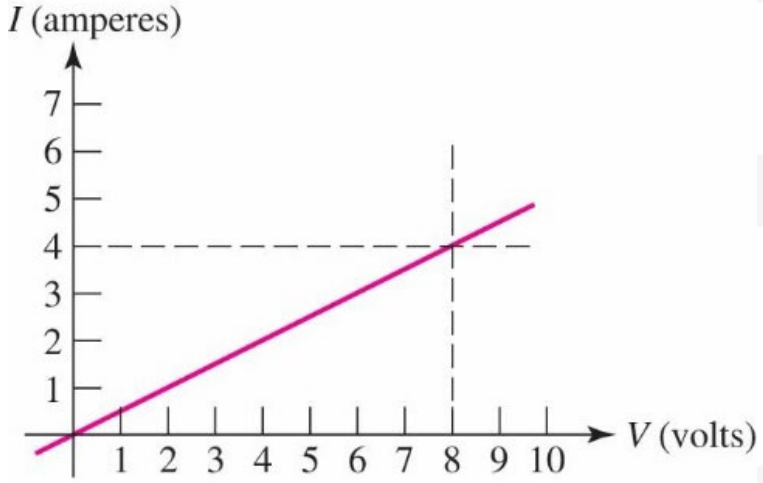
- Linearity and Superposition
- Source Transformations
- Thévenin and Norton Equivalent Circuits
- Maximum Power Transfer
- Delta-Wye Conversion

Linear Elements and Circuits

Linear element: Passive element has a linear voltage-current relationship:

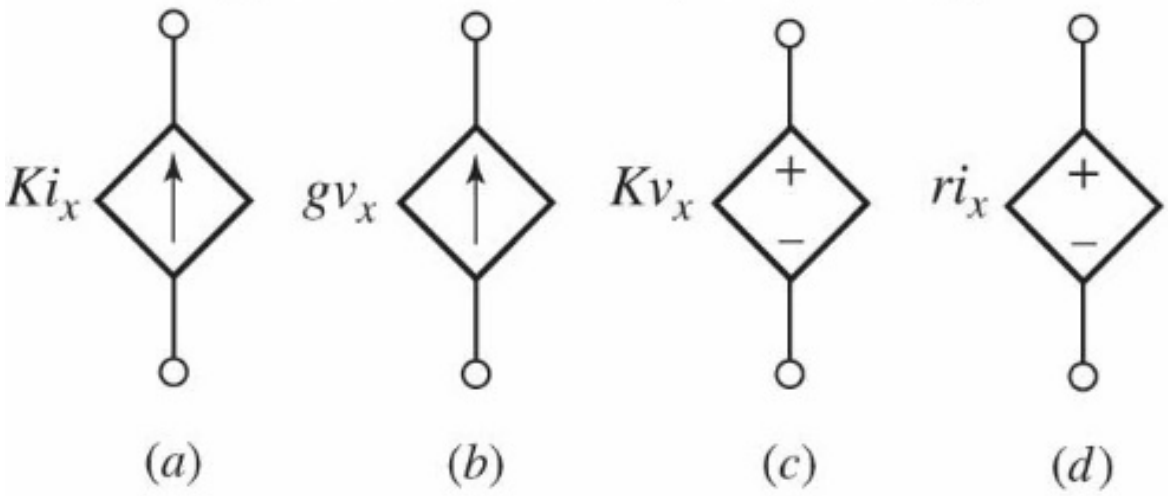
- if $i(t)$ produces $v(t)$, then $Ki(t)$ produces $Kv(t)$
- if $i_1(t)$ produces $v_1(t)$ and $i_2(t)$ produces $v_2(t)$, then $i_1(t) + i_2(t)$ produces $v_1(t) + v_2(t)$,
- *R*esistors are linear elements

$$v(t) = Ri(t)$$



Linear Elements and Circuits

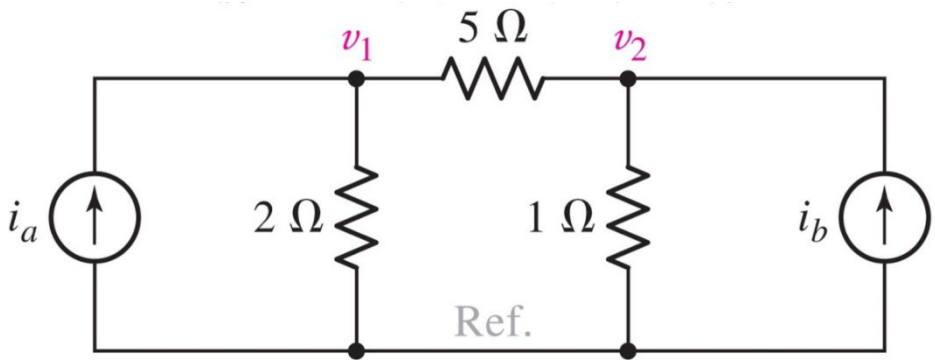
Linear dependent source: dependent current or voltage source whose output current or voltage is proportional only to the first power of a specified current or voltage variable in the circuit (or to the sum of such quantities).



Linear circuit: A circuit has only independent sources, linear dependent sources, and linear elements

The Superposition Concept 1

For the circuit shown, we have 2 independent source i_a, i_b



Question: How much of v_1 is due to source *a*, and how much is because of source *b*?

If the two sources are i_a, i_b, we get v_1, v_2

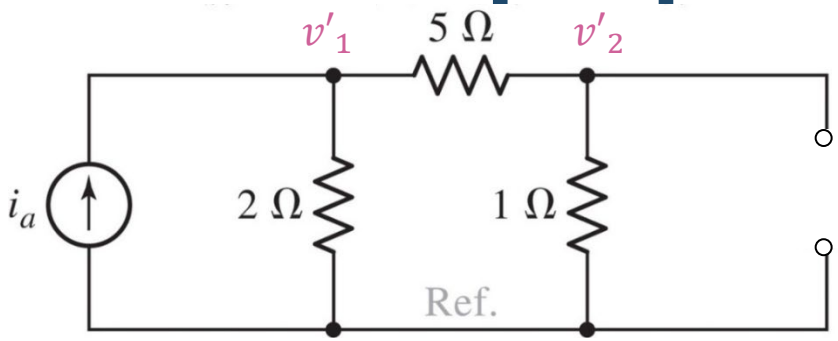
Apply KCL to node 1: $\frac{v_1 - v_2}{5} + \frac{v_1}{2} - i_a = 0$

Apply KCL to node 2: $\frac{v_2 - v_1}{5} + \frac{v_2}{1} - i_b = 0$

$$\rightarrow \begin{cases} 0.7v_1 - 0.2v_2 = i_a \\ -0.2v_1 + 1.2v_2 = i_b \end{cases}$$

Sources i_a, i_b: forcing functions;
Nodal voltages v_1, v_2: response functions(or simply responses).

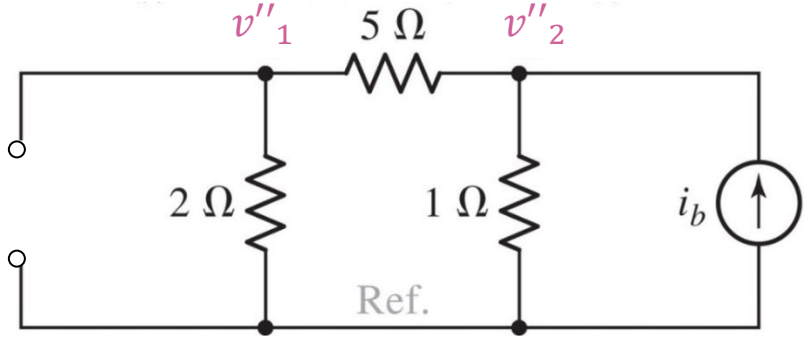
The Superposition Concept 1



If : $i_b = 0$ ➔ v'_1, v'_2

➔ $\begin{cases} 0.7v'_1 - 0.2v'_2 = i_a \\ -0.2v'_1 + 1.2v'_2 = 0 \end{cases}$

Apply KCL to node 1: $\frac{v'_1}{6} + \frac{v'_1}{2} - i_a = 0$



If : $i_a = 0$ ➔ v''_1, v''_2

Apply KCL to node 2: $\frac{v''_2}{1} + \frac{v''_2}{7} - i_b = 0$

Question: How much of v_1 is due to source *a*, and how much is because of source *b*?

$v_1 = v'_1 + v''_1$

The Superposition Theorem

In a linear network, the **voltage across** or the **current through** any element may be calculated by *adding algebraically* all the individual voltages or currents caused by the separate independent sources acting "alone", that is, with

- all other independent voltage sources replaced by short circuits
- all other independent current sources replaced by open circuits

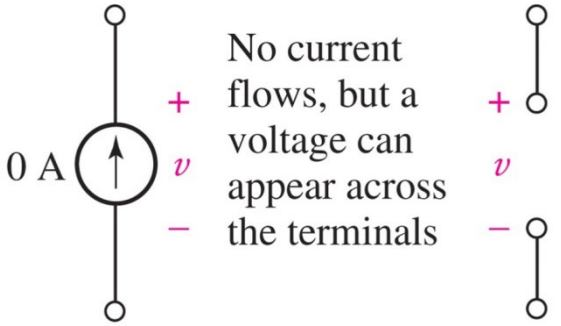


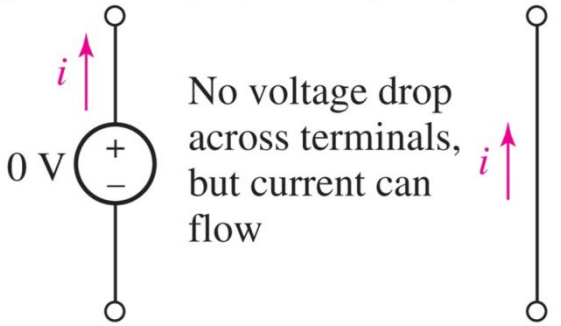
Applying Superposition

Leave one source ON and turn all other sources OFF:

- current sources: *set i*=0.
- These become *open circuits.*
- voltage sources: *set v*=0.
- These become *short circuits.*
- *Find the response from this source.*

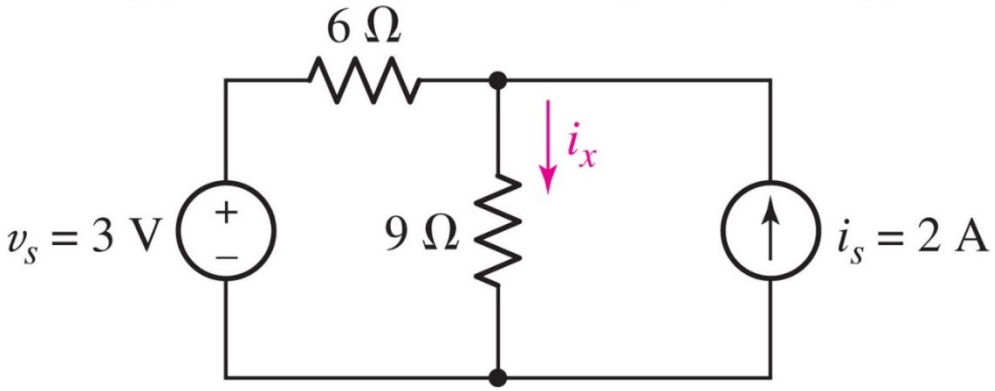
Add the resulting responses to find the total response.



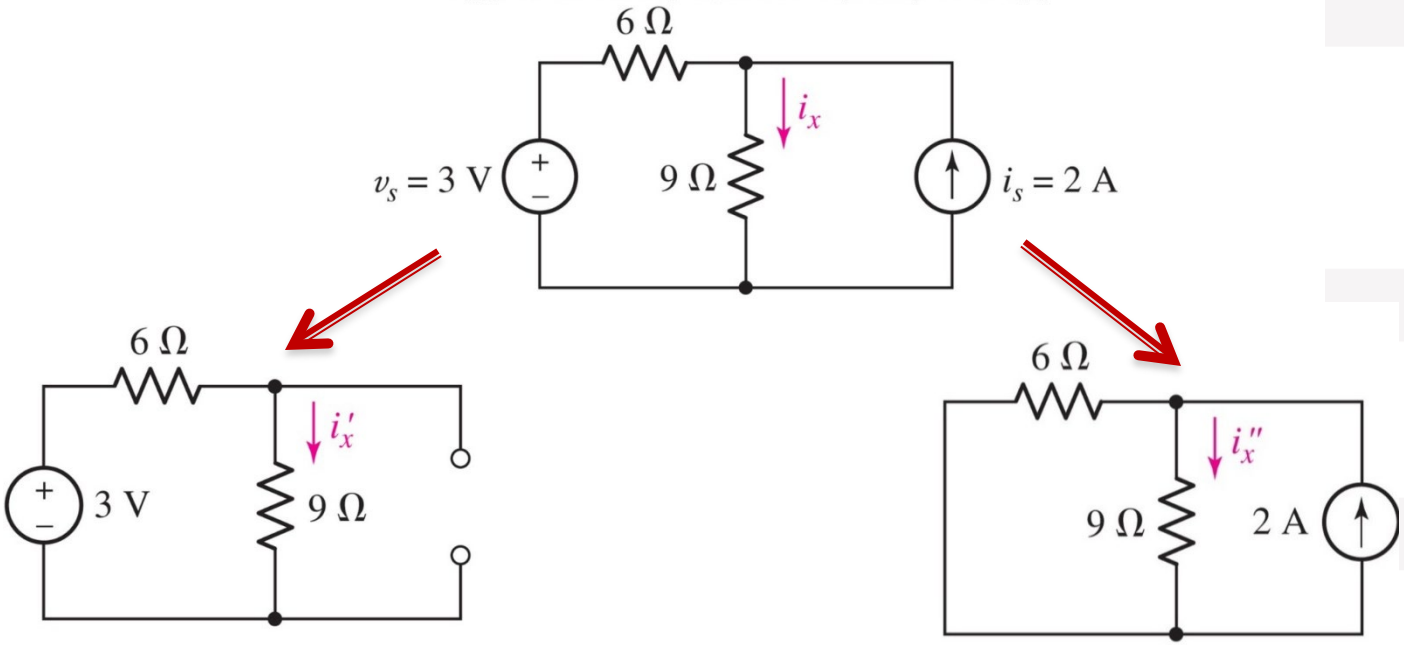


Superposition Example

Use superposition to solve for the current i_x



Superposition Example



First, turn the current source off:

$$i_x' = \frac{3}{6+9} = 0.2\,\text{A}$$

Then, turn the voltage source off:

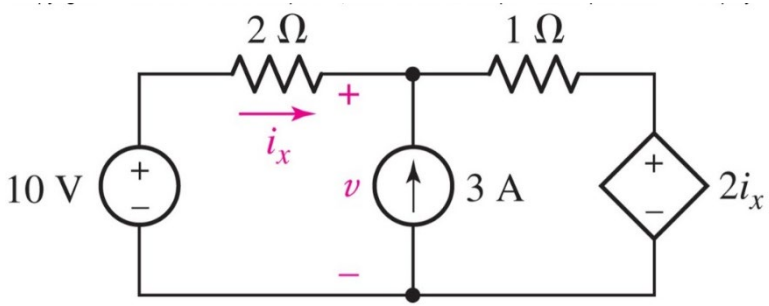
$$i_x'' = \frac{6}{6+9}(2) = 0.8\,\text{A}$$

Finally, combine the results:

$$i_x = i_x' + i_x'' = 0.2 + 0.8 = 1.0\,\text{A}$$

Superposition with a Dependent Source 1

Use superposition to solve for the current i_x



Nodal analysis:

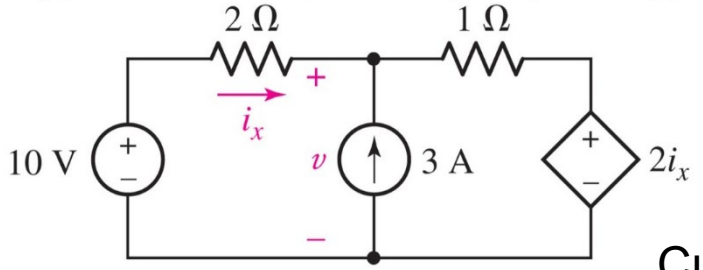
Apply KCL at node 1: $\frac{v_1-10}{2}+\frac{v_1-2i_x}{1}=3$

Relate i_x with v_1 : $i_x=\frac{10-v_1}{2}$

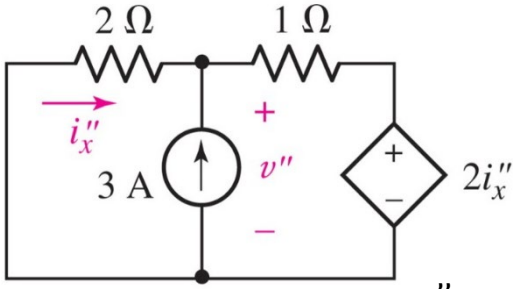
Solve: $v=v_1=7.2$ V, $i_x=1.4$ A

Superposition with a Dependent Source 2

Use superposition to solve for the current i_x



Voltage source off

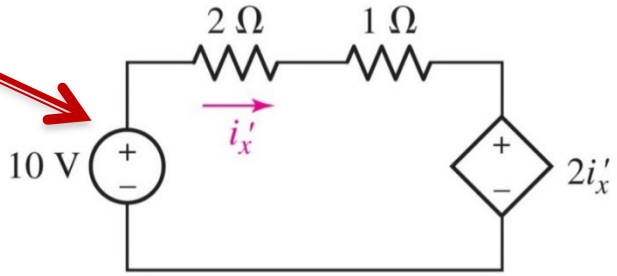


Apply KCL at node 1: $\frac{v''}{2} + \frac{v''-2i''_x}{1} = 3$

Relate i''_x with v'' : $i''_x = \frac{-v''}{2}$

Solve: $i''_x = -0.6$ A

Current source off



$-10 + 2i'_x + i'_x + 2i'_x = 0$

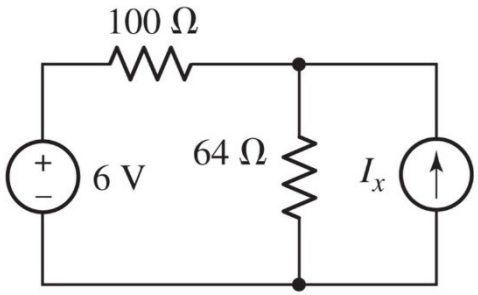
$i'_x = 2$ A

$$i_x = i_x' + i_x'' = 2 + (-0.6) = 1.4 \text{ A}$$

When applying superposition to circuits with *dependent* sources, these *dependent* sources are never "turned off."

Example: Power Ratings

Each resistor is rated to a maximum of 250 mW. Determine the maximum *positive* current to which the source I_x can be set before any resistor exceeds its power rating.

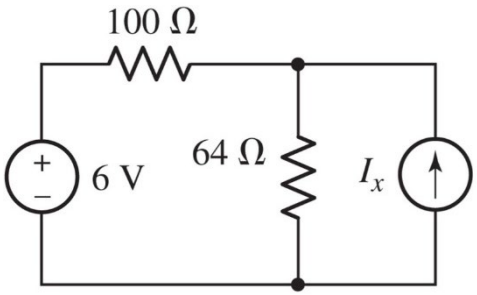


The maximum current of each resistor

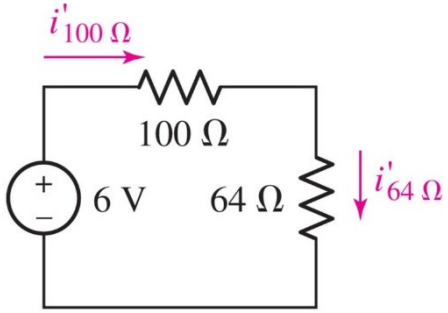
$$i_{max} = \sqrt{\frac{P_{max}}{R}}\,, \quad i_{100\Omega} < 50mA,\ i_{64\Omega} < 62.5mA$$

$$V_{max} = \sqrt{P_{max}R}\,, \quad V_{100\Omega} < 5\text{V},\ V_{64\Omega} < 4\text{V}$$

Example: Power Ratings

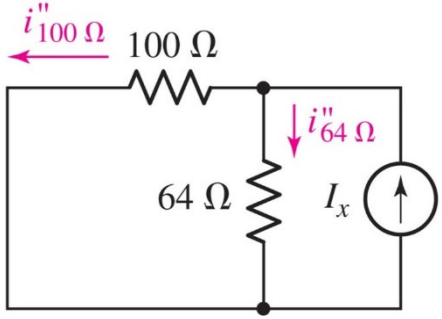


Current source off



$$i'_{100\Omega} = i'_{64\Omega} = 36.59mA$$

Voltage source off



$$i''_{100\Omega} = \frac{64}{100+64} I_x = 0.39 I_x$$

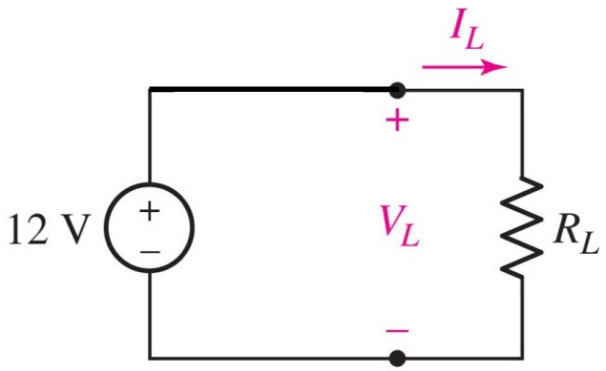
$$i''_{64\Omega} = \frac{100}{100+64} I_x = 0.61 I_x$$

$$|i''_{100\Omega} - i'_{100\Omega}| < 50mA$$
$$i'_{64\Omega} + i''_{64\Omega} < 62.5mA$$

$$Answer: I_x < 42.49 \text{ mA}$$

Practical Voltage Sources

Ideal voltage sources: a first approximation model for a battery.



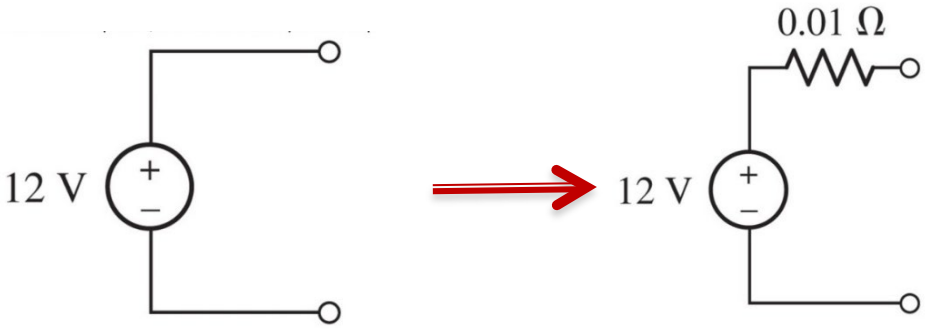
If R_L=1Ω, I_L=?

If R_L=1μΩ, I_L=?

If R_L=0, I_L=?

Why do real batteries have a current limit and experience voltage drop as current increases?

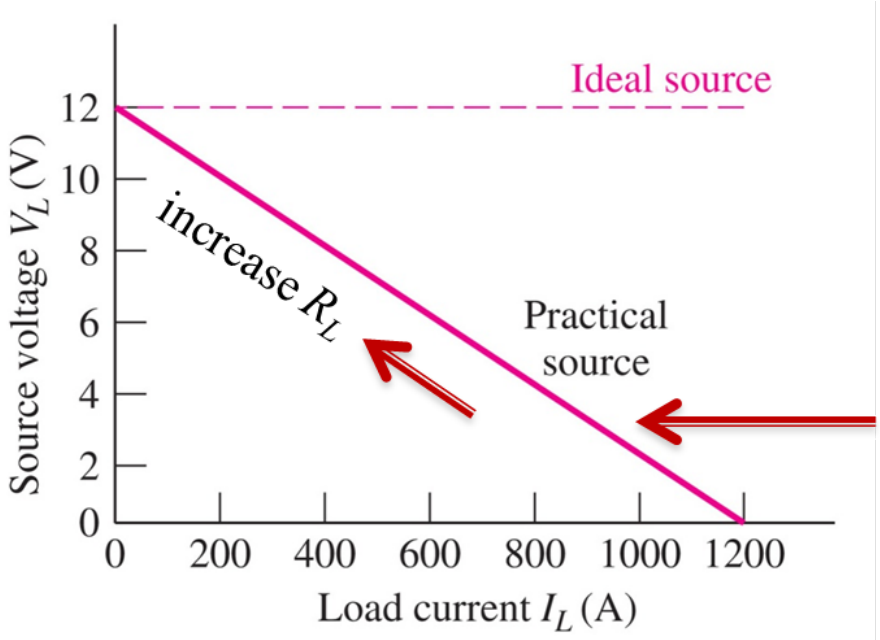
Two battery models:

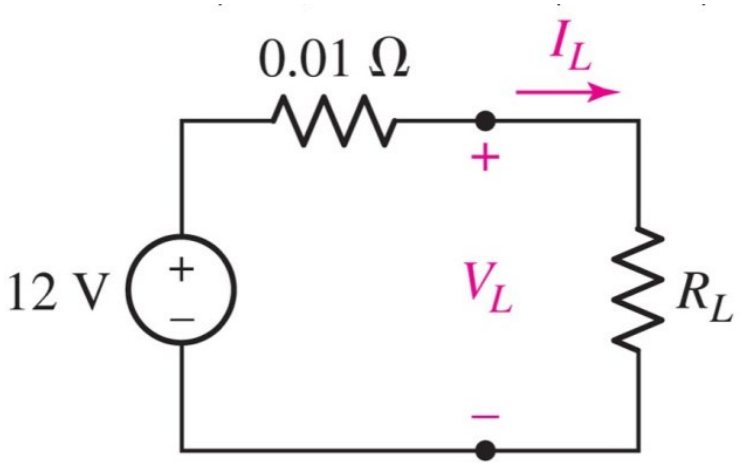


Practical Source: Effect of Connecting a Load

The practical voltage source model:

$$V_L = 12 - 0.01\, I_L$$

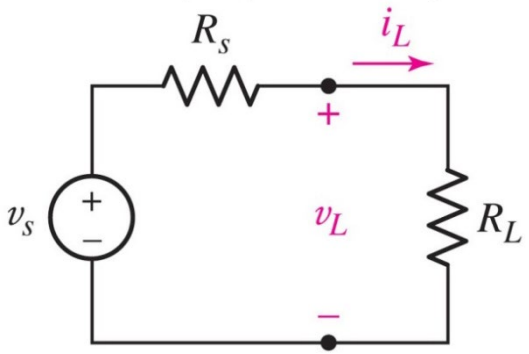




This line represents all possible R_L

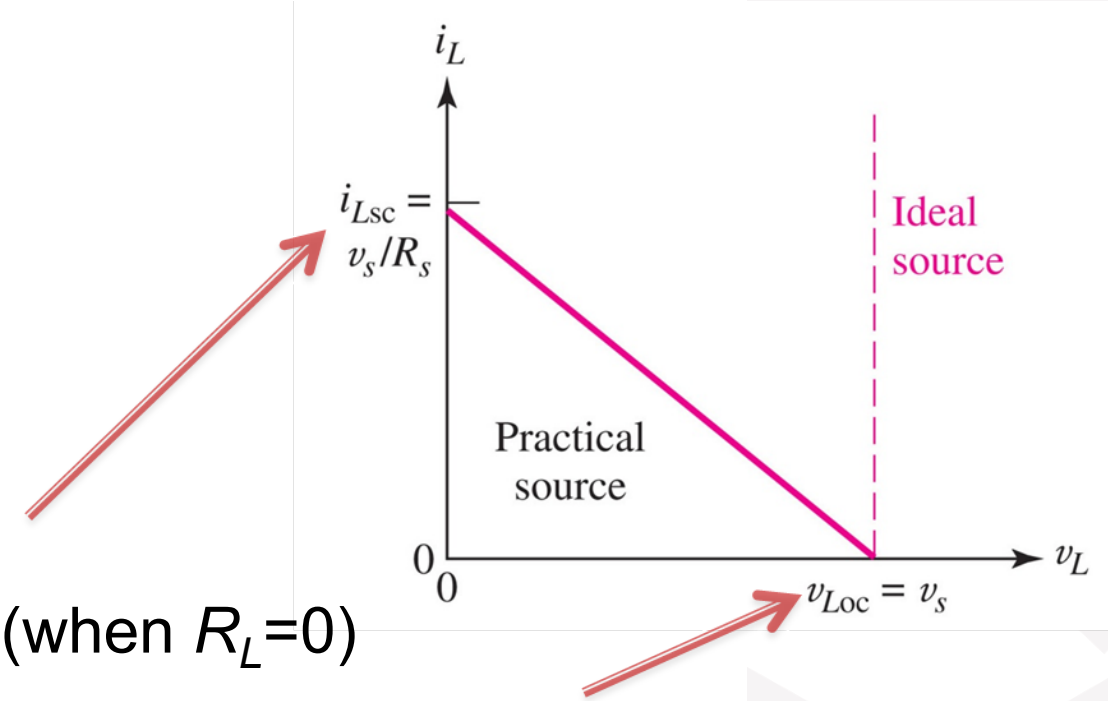
Practical Voltage Source

The source has an internal resistance or output resistance, which is modeled as R_s



The linear relationship between v_L and i_L:

$$v_L = v_s - R_s i_L$$

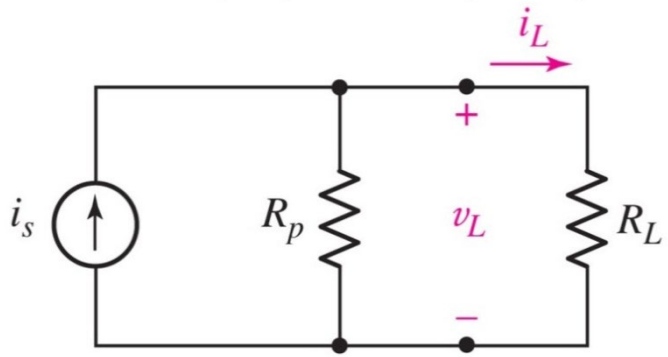


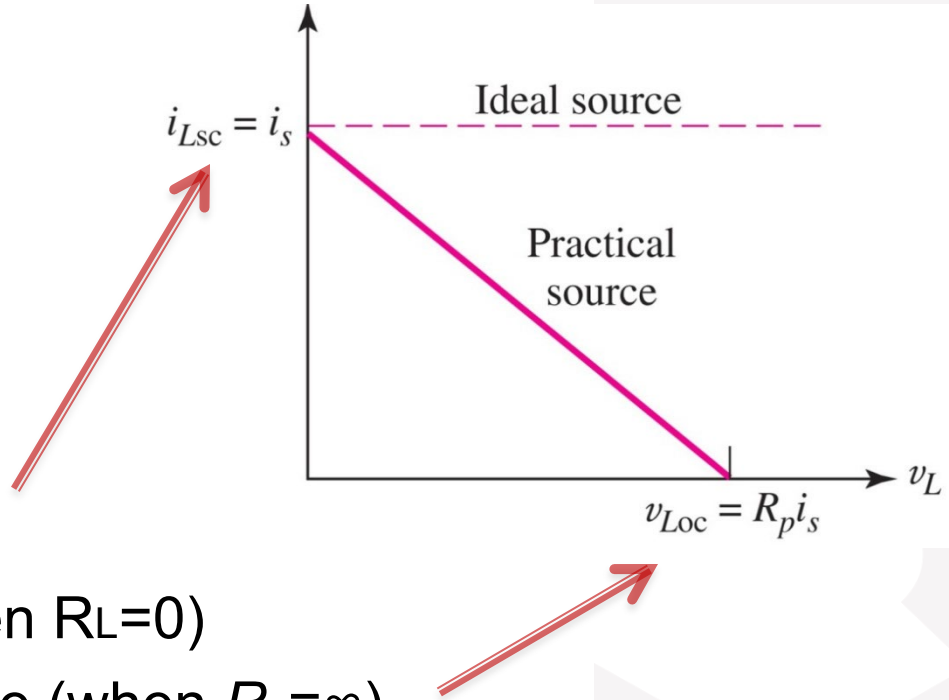
short circuit current (when R_L=0)

open circuit voltage (when R_L=∞)

Practical Current Source

The source has an internal *parallel* resistance which is modeled as R_p





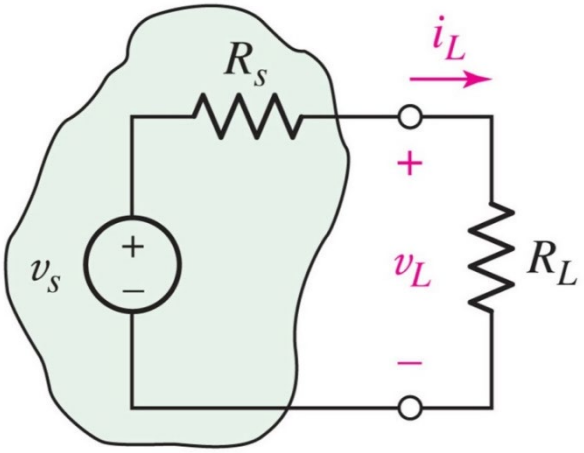
The linear relationship between v_L and i_L:

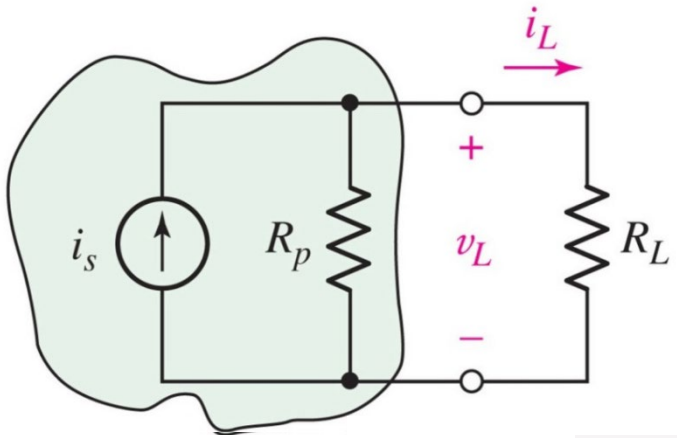
$$i_L = i_s - \frac{v_L}{R_p}$$

short circuit current (when R_L=0)

open circuit voltage (when R_L=∞)

Source Transformation and Equivalent Sources





$$v_L = v_s \frac{R_L}{R_s + R_L}$$

$$v_L = i_s \frac{R_p}{R_p + R_L} \cdot R_L$$

The sources are equivalent if

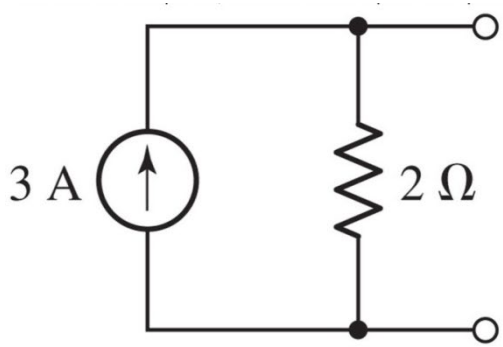
$$R_s = R_p \text{ and } v_s = i_s R_s$$

Source Transformation

The circuits (a) and (b) are equivalent at the terminals.

$$R_s = R_p = 2\Omega$$

$$v_s = i_s R_s = 3A * 2\Omega = 6\text{V}$$

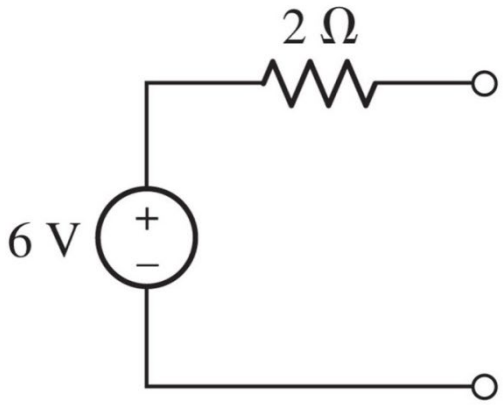


(a)

If given circuit (a), but circuit (b) is more convenient, switch them!

This process is called

source transformation.



(b)

The head of the current source arrow corresponds to the "+" terminal of the voltage source.

Source Transformation

If $R_L = 4\Omega$:

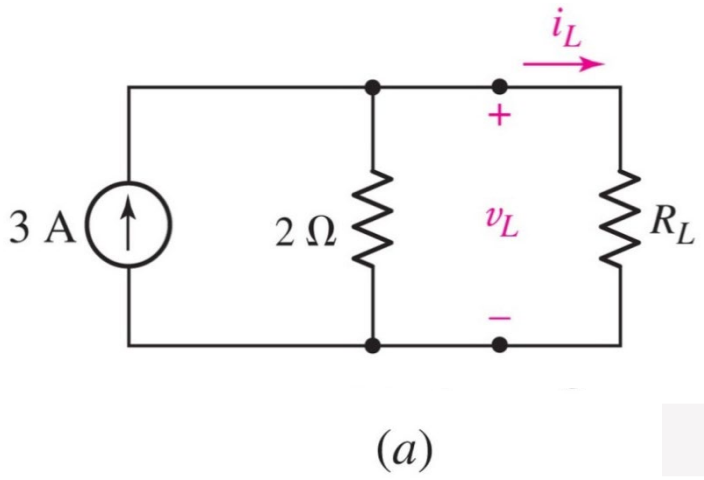
(a) Load terminal:

$$i_L = \frac{R_s}{R_s+R_L} i_s = 1A,\ V_L = i_L R_L = 4V,$$

$$P_{R_L} = i_L v_L = 4W$$

Inside the practice source:

$$P_{3A} = -3\text{A} * v_L = -12W,\ P_{R_p} = {v_L^2}/{R_p} = 8W$$



(a)

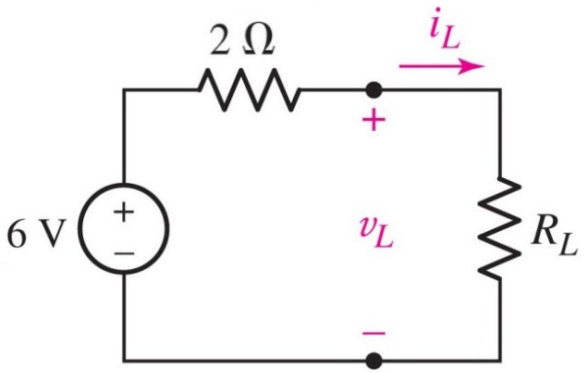
(b) Load terminal:

$$i_L = \frac{v_s}{R_s+R_L} = 1A,\ V_L = i_L R_L = 4V,$$

$$P_{R_L} = i_L v_L = 4W$$

Inside the practice source:

$$P_{6V} = -6V * i_L = -6W\ ,\ P_{R_s} = i_L^2 * R_s = 2W$$

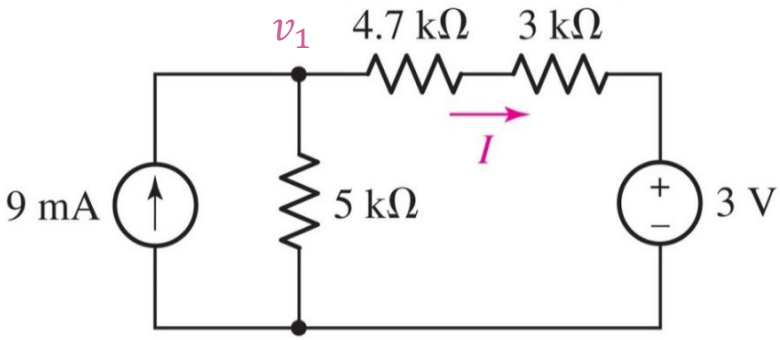


(b)

The two practical sources are equivalent only with respect what transpires at the load terminals; they are not equivalent internally!

Example: SourceTransformation

Calculate the current *I* in the circuit below:



Method 1:

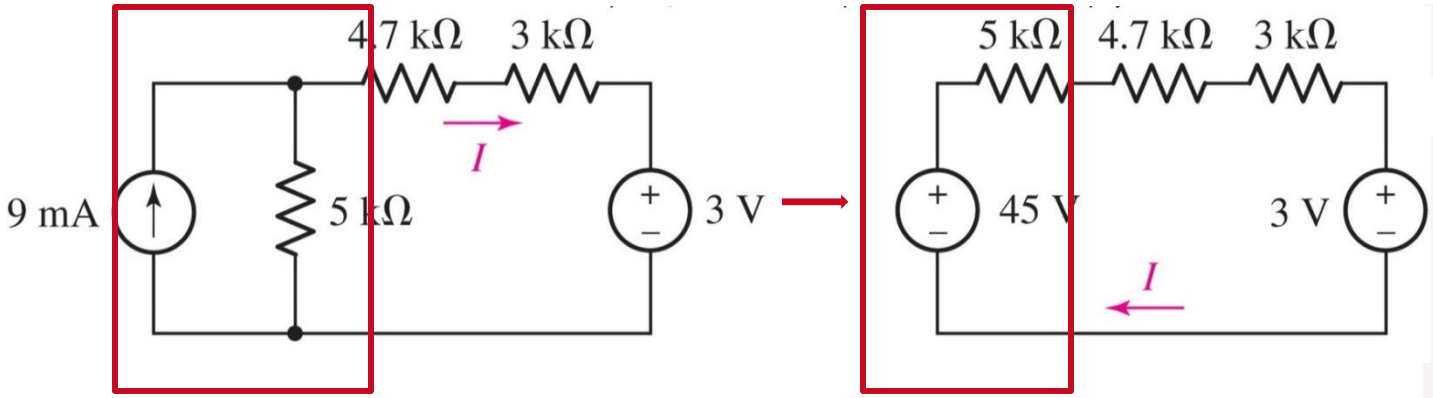
Apply KCL: $9mA - \frac{v_1}{5k\Omega} - \frac{v_1 - 3V}{4.7k\Omega + 3k\Omega} = 0$

$$v_1 = 28.465\ V$$

$$I = \frac{v_1 - 3V}{7.7k\Omega} = 3.307mA$$

Example: SourceTransformation

Calculate the current *I* in the circuit below using source transformation



Method 2: source transformation

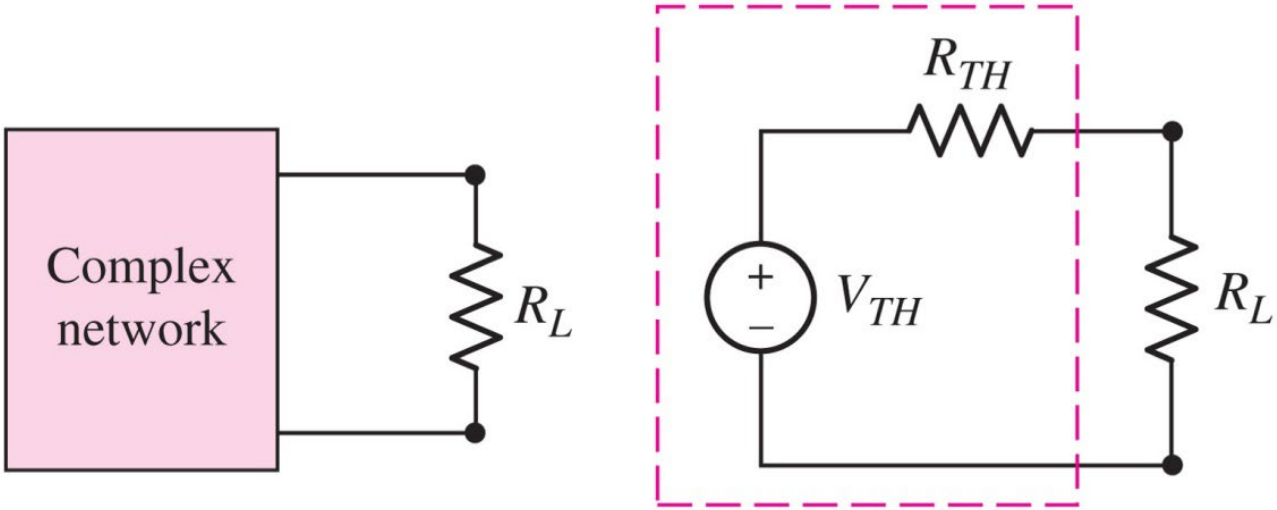
Equivalent voltage source:

$$R_s = R_p = 5k\Omega,\ V_s = i_s R_s = 45V$$

$$I = (45-3)/(5+4.7+3) = 3.307\ \text{mA}$$

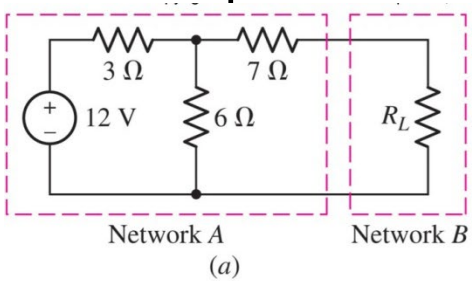
Thévenin Equivalent Circuits

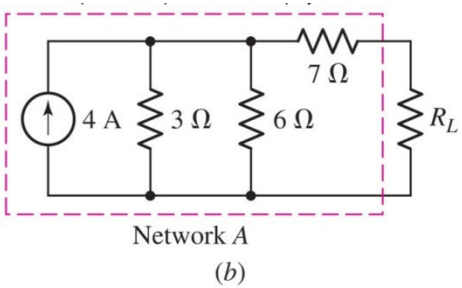
Thévenin's theorem: a linear network can be replaced by its Thévenin equivalent circuit, as shown below:



Thévenin Equivalent using Source Transformation

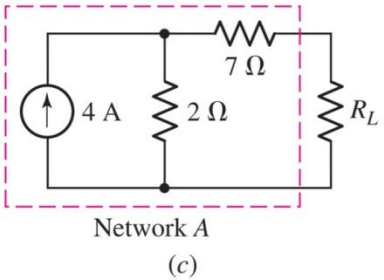
We can repeatedly apply source transformation on network A to find its Thévenin equivalent circuit.

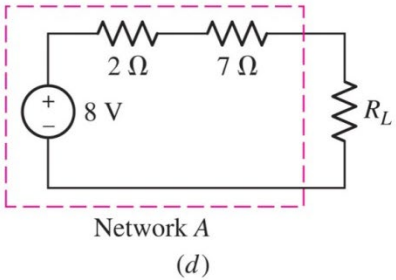




(a) to (b) Source transformation:

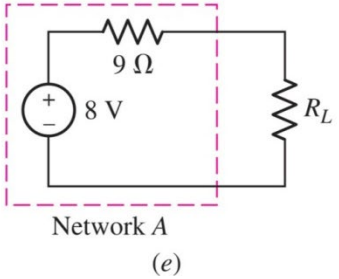
$$R_p = R_s = 3\Omega, i_s = \frac{v_s}{R_s} = 4A$$





(c) to (d) Source transformation:

$$R_s = R_p = 2\Omega, v_s = i_s R_s = 8V$$



This method has limitations- not all circuits can be source transformed.

Finding the Thévenin Equivalent

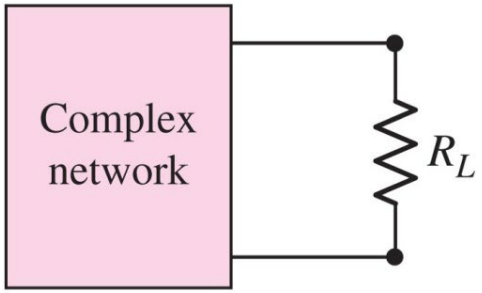
Step1: Disconnect the load ($R_L = \infty$). Find the open circuit voltage v_{oc}

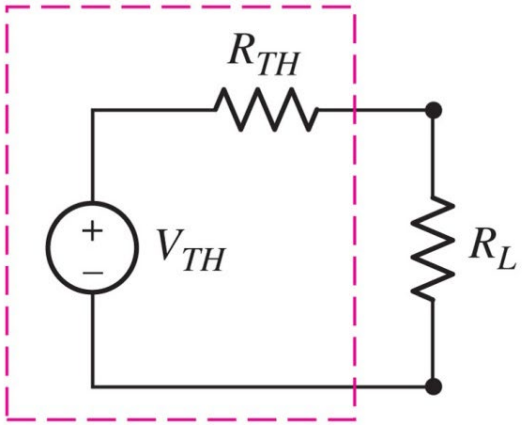
Step2: Find the equivalent resistance R_{eq} of the network with all independent sources turned off.

Then:

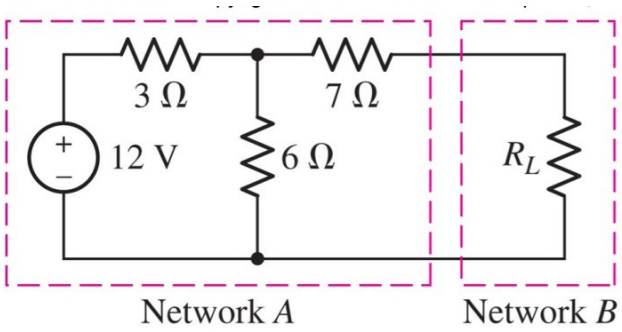
$$V_{TH} = v_{oc} \text{ and}$$

$$R_{TH} = R_{eq}$$



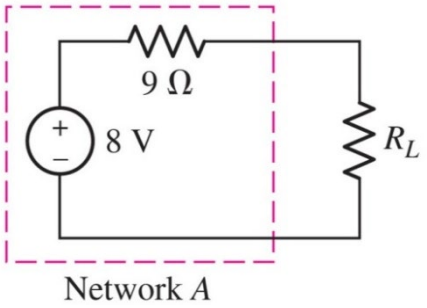


Thévenin Example

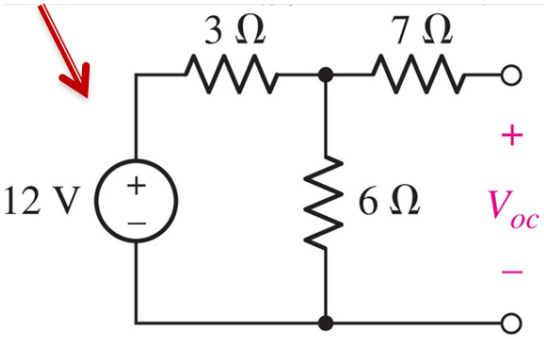


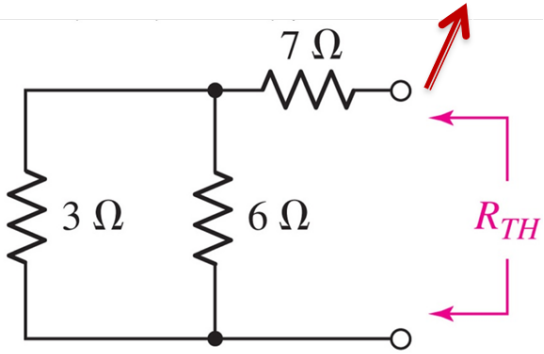
$$V_{TH} = v_{oc} \text{ and}$$

$$R_{TH} = R_{eq}$$



Disconnect the load ($R_L = \infty$). Find the open circuit voltage v_{oc}



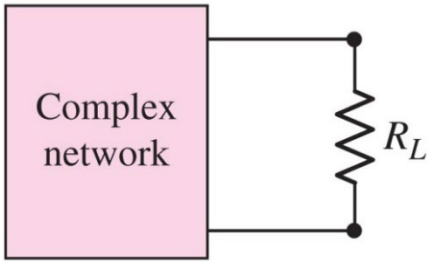


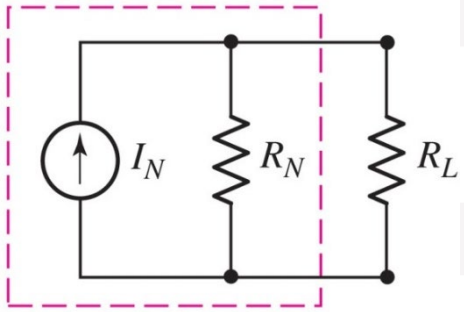
Find R_{eq} of the network with all independent sources turned off.

$$Voc = 12\frac{6}{3+6} = 8V \qquad R_{TH} = 3||6 + 7 = 9\Omega$$

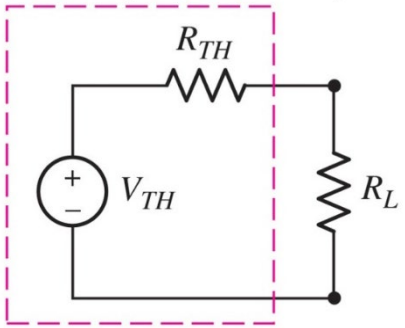
Norton Equivalent Circuits

Norton's theorem: a linear network can be replaced by its Norton equivalent circuit, as shown below:





The Thévenin and Norton equivalents are source transformations of each other!



$$R_{TH} = R_N = R_{eq} \text{ and } v_{TH} = i_N R_{eq}$$

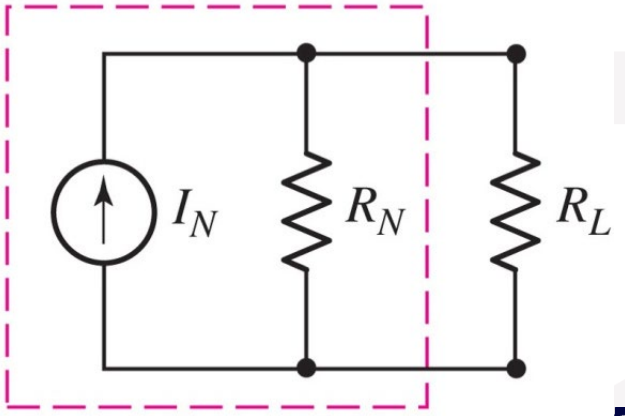
Finding the Norton Equivalent

Step1: Replace the load with a short circuit. ($R_L = 0$) Find the short circuit current i_{sc}

Step2: Disconnect the load ($R_L = \infty$). Find the equivalent resistance R_{eq} of the network with all independent sources turned off.

Then:

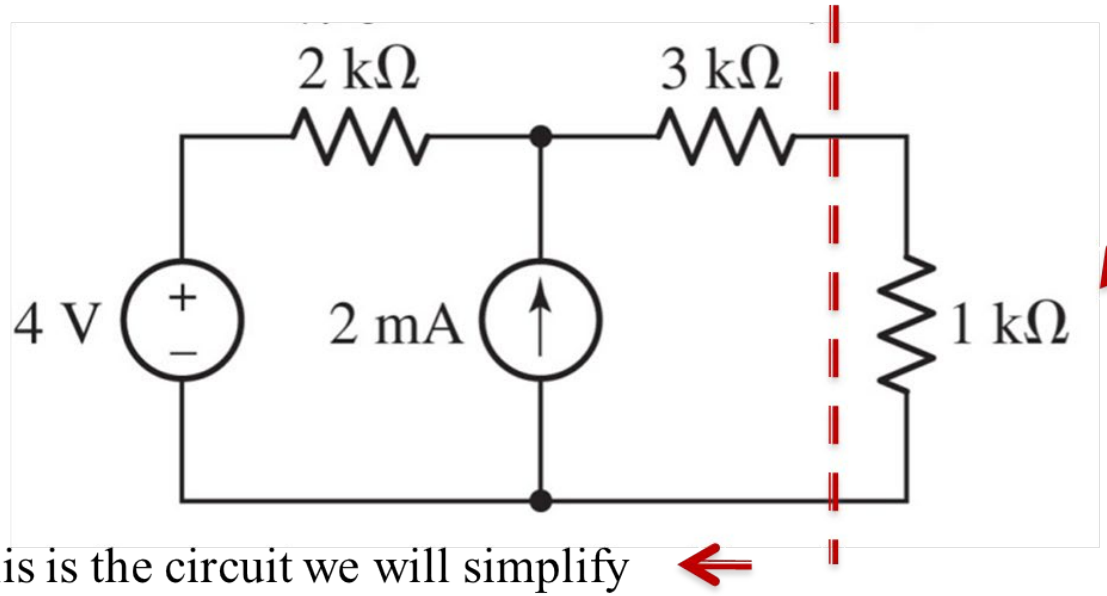
$$I_N = i_{sc} \text{ and } R_N = R_{eq}$$



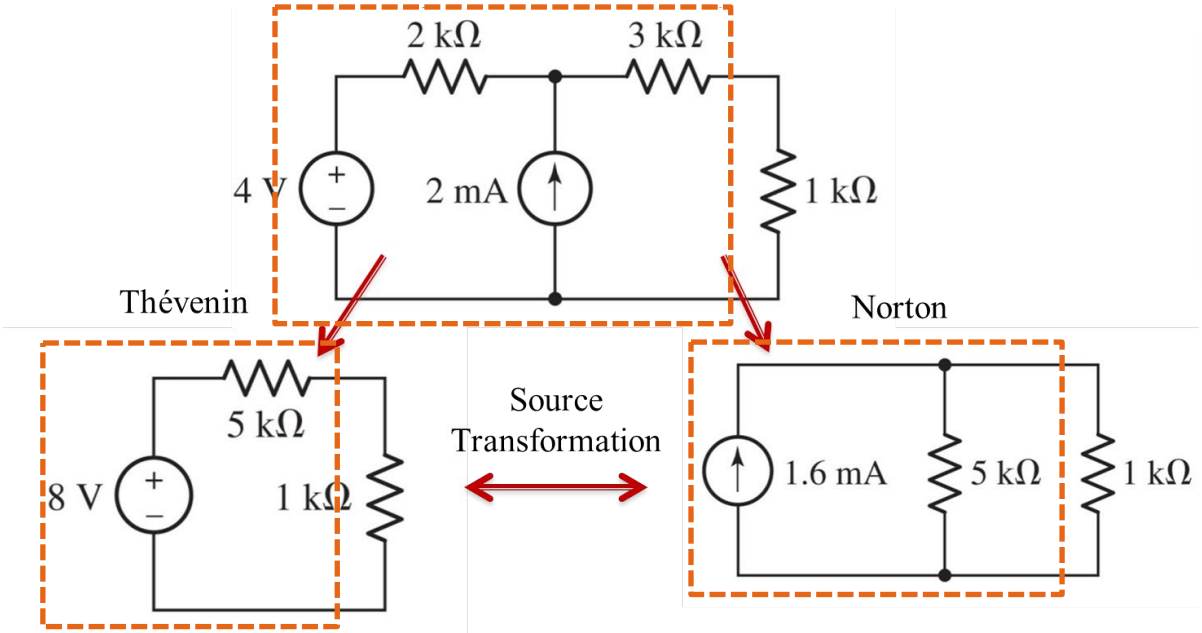
Example: Norton and Thévenin 1

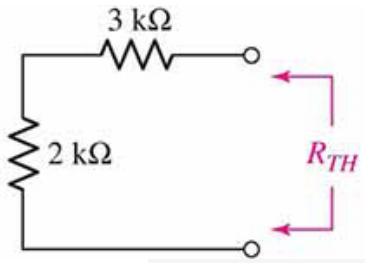
Find the Thévenin and Norton equivalents for the network faced by the 1 kΩ resistor.

Answer: next slide



Example: Thévenin and Norton 2





Thévenin Equivalent:

$$V_{TH} = V_{oc} = 8V$$

$$R_{TH} = 5k\Omega$$

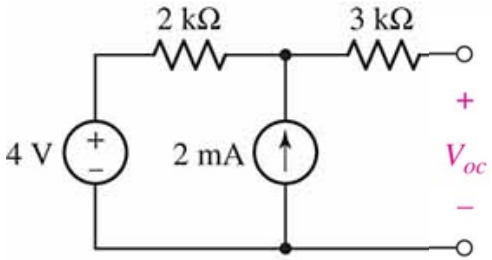
Source Transformation:

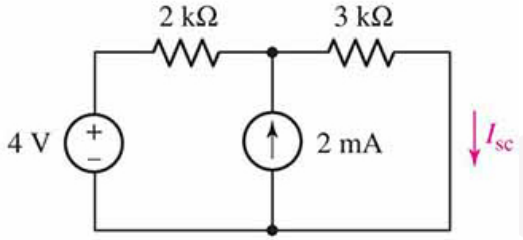
$$R_p = R_s = 5k\Omega$$

$$i_s = \frac{v_s}{R_s} = 1.6mA$$

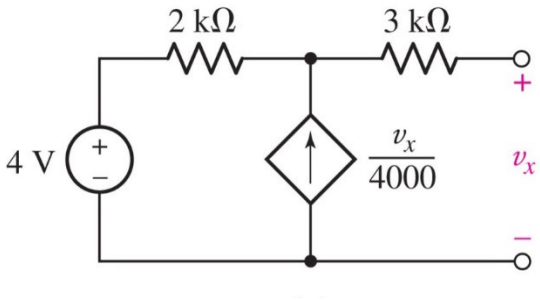
Norton Equivalent:

$$I_N = I_{sc} = 1.6mA$$





Thévenin Example: Handling Dependent Sources 1

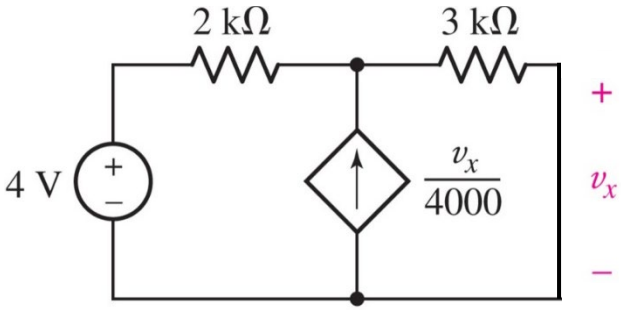


Thévenin:

Apply KVL: $-4 + (-\frac{v_x}{4000}) * 2000 + v_x = 0$

$V_{oc} = v_x = 8V$

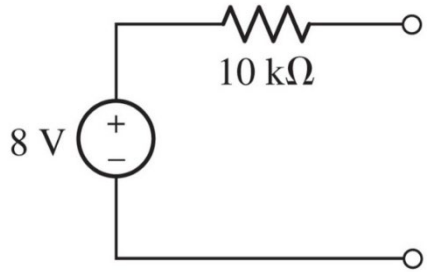
$R_{TH} = ?$



Norton:

$$v_x = 0, I_{sc} = \frac{4}{2000 + 3000} = 0.8mA$$

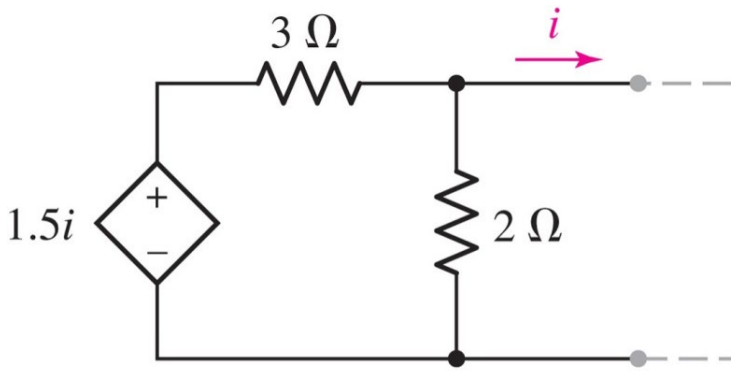
$$R_{TH} = \frac{V_{oc}}{I_{sc}} = 10k\Omega$$

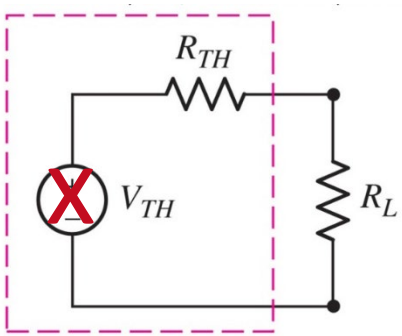


One method to find the Thévenin equivalent of a circuit with a dependent source: find V_{TH} and I_N and solve for $R_{TH} = V_{TH} / I_N$

Thévenin Example: Handling Dependent Sources 2

Finding the ratio V_{TH} / I_N fails when both quantities are zero





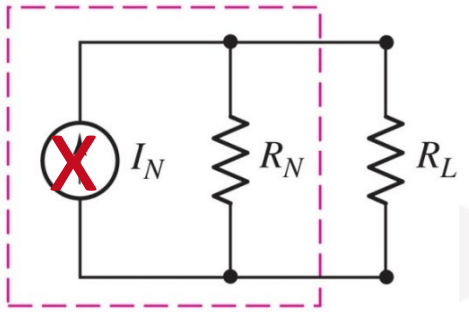
$R_{TH} = ?$

Open circuit:

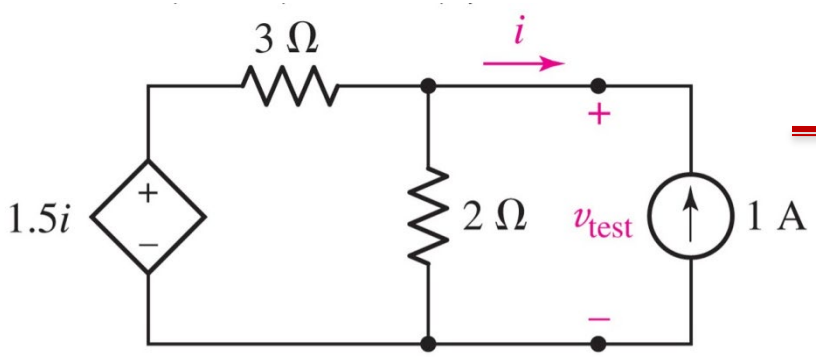
$i = 0 \longrightarrow V_{oc} = 0$

Short circuit:

$-1.5i + 3i = 0 \rightarrow I_{sc} = i = 0$



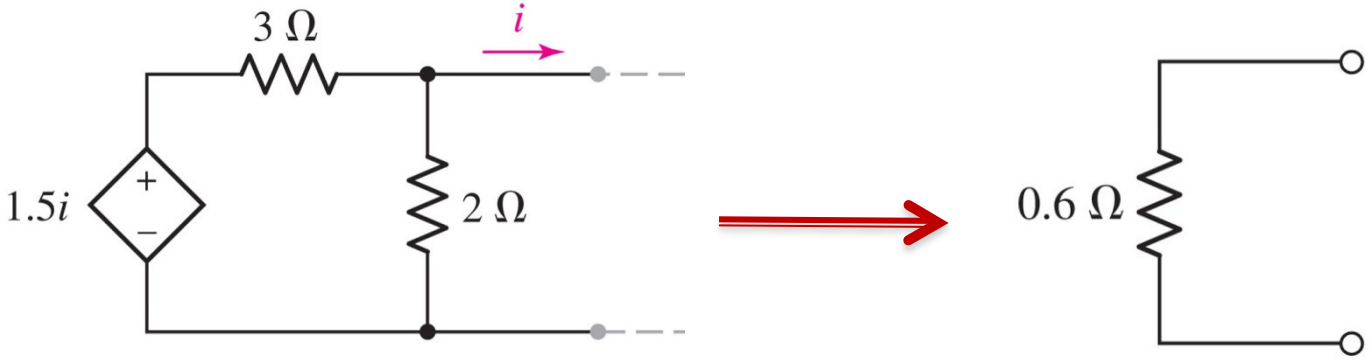
Thévenin Example: Handling Dependent Sources 3



$$\frac{v_{test}}{2} + \frac{v_{test} - (1.5i)}{3} = 1$$

$$i = -1$$

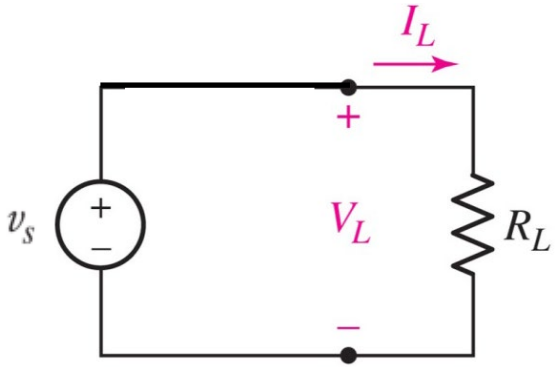
Solve: $v_{test} = 0.6$ V, and so $R_{TH} = 0.6\ \Omega$

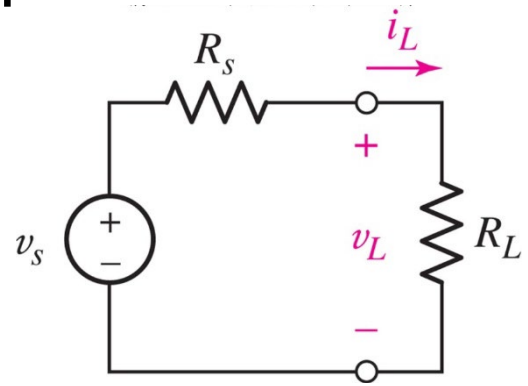


One method to find the Thévenin equivalent of a circuit with a dependent source: apply a test source

Maximum Power Transfer

What load resistor will allow the **practical source** to deliver the maximum power to the load?





Idear source:

$$p_L = \frac{v_s^2}{R_L}$$

Pratical source:

$$p_L = i_L^2 R_L = \frac{v_s^2 R_L}{(R_s + R_L)^2}$$

[solve $dp_L/dR_L = 0$, check $R_L = 0$ & $R_L = \infty$]

Answer: $R_L = R_s$

$$p_{\text{max|delivered to load}} = \frac{v_s^2}{4R_s}$$

Maximum Power Transfer

Maximum power transfer theorem:

An independent voltage source in series with a resistance Rs (or an independent current source in parallel with a resistance Rs) delivers maximum power to a load resistance R_L such that R_L = Rs.

An alternative expression:(In terms of the Thévenin equivalent resistance of a network):

A network delivers maximum power to a load resistance R_L when R_L is equal to the Thévenin equivalent resistance of the network (R_{TH}).

$$p_{\max}|_{\text{delivered to load}} = \frac{v_s^2}{4R_s} = \frac{v_{TH}^2}{4R_{TH}}$$



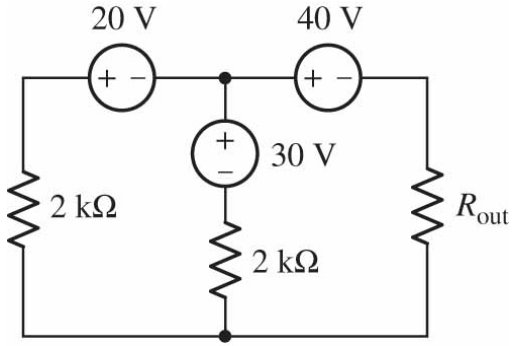
Example: Maximum Power Transfer

Consider the circuit below:

(a) What is the maximum power that can be delivered to Rout?

(b) If Rout = 3 kΩ, find the power delivered to it.

(c) What two different values of R_{out} will have exactly 20 mW delivered to them?

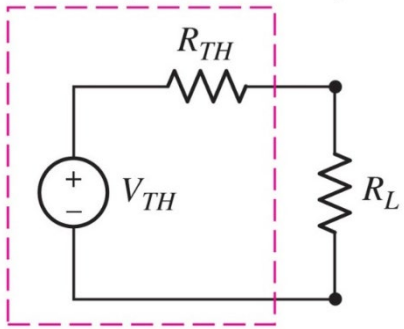


Open circuit:

In single loop: $\frac{v+20}{2_{\mathrm{k}}}+\frac{v-30}{2\mathrm{k}}=0$

$V_{TH}=V_{oc}=v-40=-35V$

$R_{TH}=1k\Omega$



(a)When $R_{out}=R_{TH}$ the load get the maximum power

$p_{max}=(\frac{V_{TH}}{R_{TH}+R_{out}})^2R_{out}=306mW$

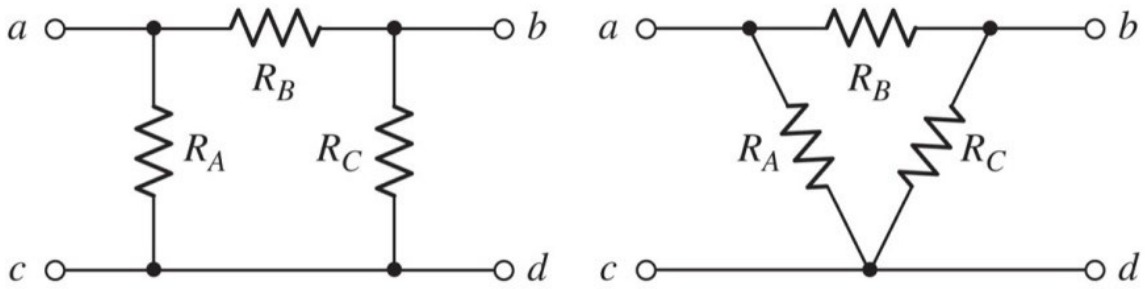
(b) $p=(\frac{V_{TH}}{R_{TH}+R_{out}})^2R_{out}=230mW$

(c)$(\frac{V_{TH}}{R_{TH}+R_{out}})^2R_{out}=20mW$

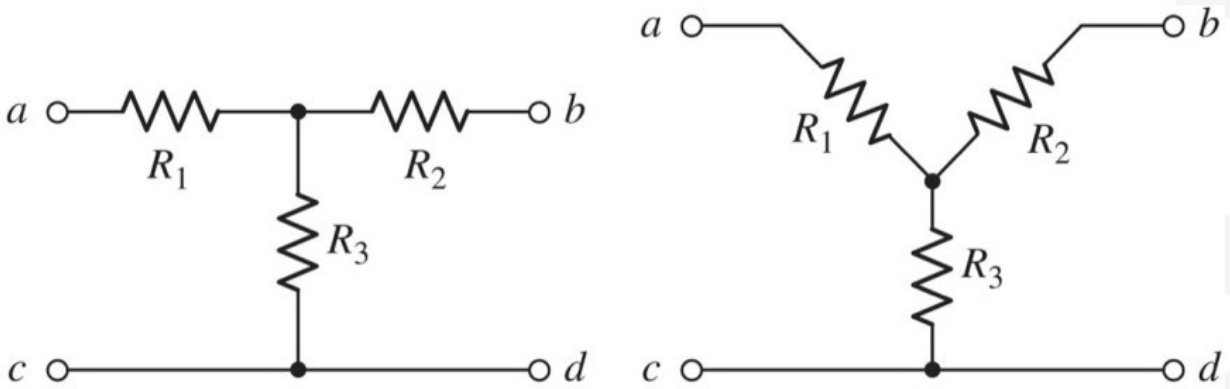
$R_{out1}=59.2k\Omega, R_{out2}=16.88\Omega$

Δ-Y (delta-wye) Conversion[1]

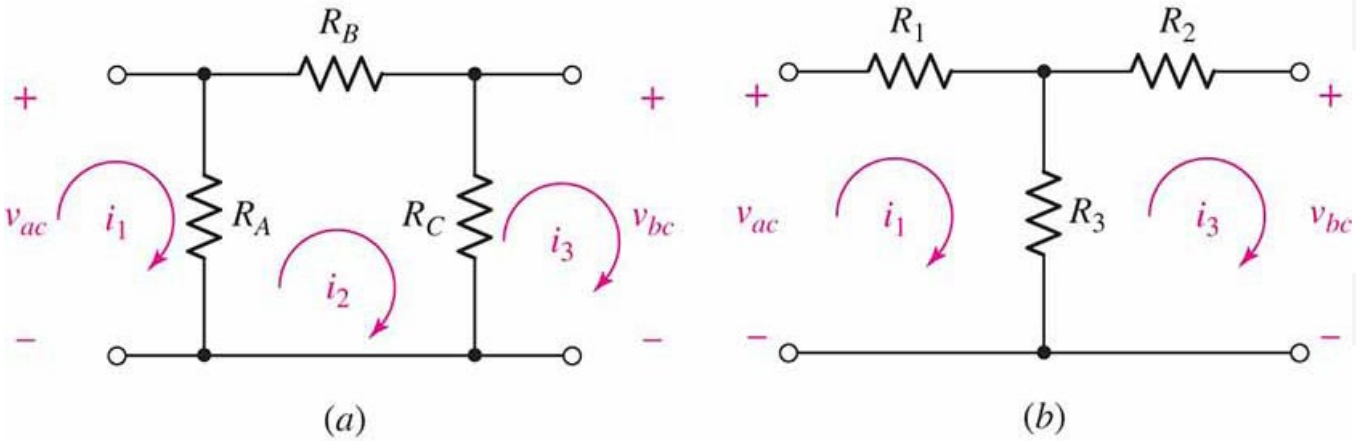
The following resistors form a Δ:



The following resistors form a Y:



Δ-Y (delta-wye) Conversion 2



If the two networks are equivalent, then the terminal voltages and currents must be equal.

$$\begin{aligned} R_A i_1 - R_A i_2 &= v_{ac} \\ -R_A i_1 + (R_A + R_B + R_C) i_2 - R_C i_3 &= 0 \\ -R_C i_2 + R_C i_3 &= -v_{bc} \end{aligned}$$

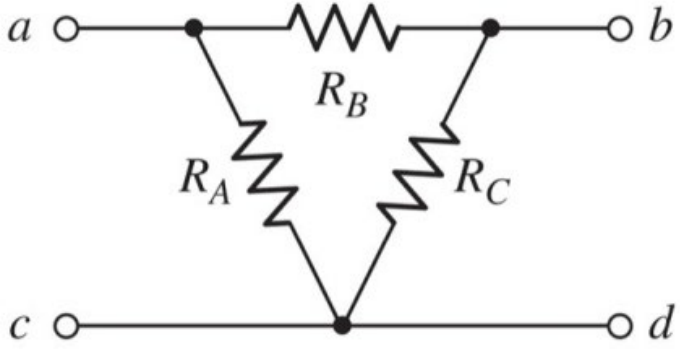
$$\begin{aligned} (R_1 + R_3) i_1 - R_3 i_3 &= v_{ac} \\ -R_3 i_1 + (R_2 + R_3) i_3 &= -v_{bc} \end{aligned}$$

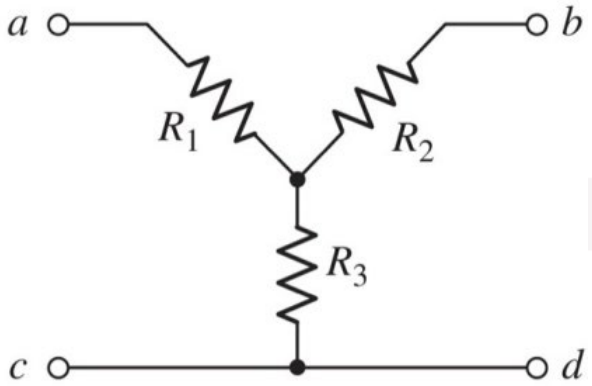
Remove i_2

$$\left(R_A - \frac{R_A^2}{R_A + R_B + R_C}\right) i_1 - \frac{R_A R_C}{R_A + R_B + R_C} i_3 = v_{ac}$$

$$-\frac{R_A R_C}{R_A + R_B + R_C} i_1 + \left(R_C - \frac{R_C^2}{R_A + R_B + R_C}\right) i_3 = -v_{bc}$$

Δ-Y (delta-wye) Conversion[3]





This Δ is equivalent to the Y if

$$R_A = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}$$

$$R_B = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}$$

$$R_C = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}$$

This Y is equivalent to the Δ if

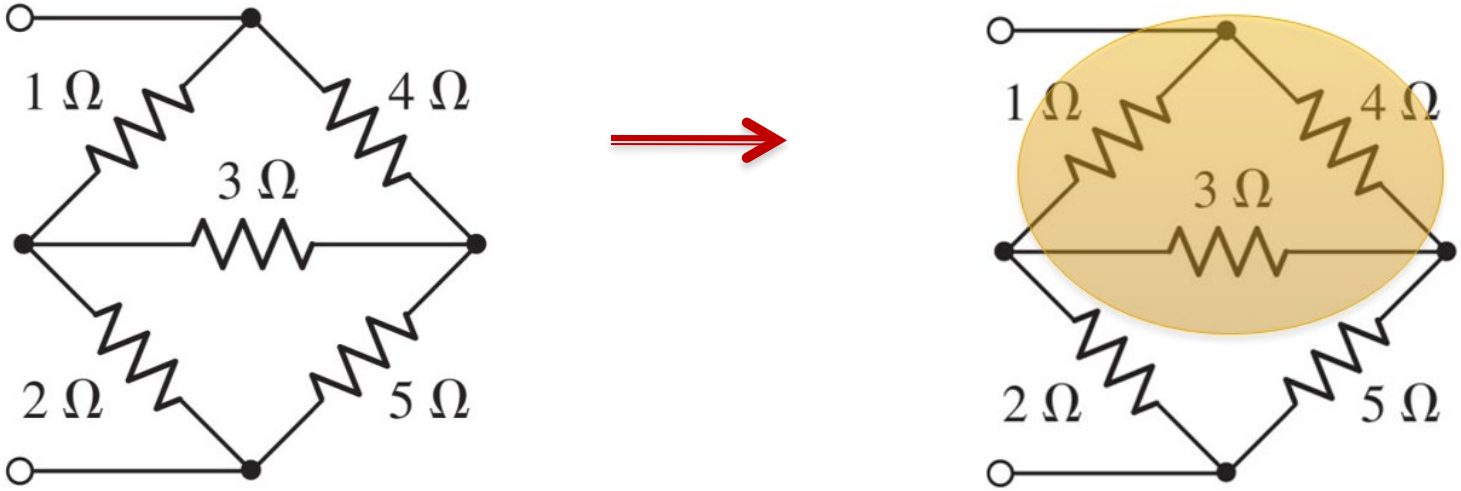
$$R_1 = \frac{R_AR_B}{R_A + R_B + R_C}$$

$$R_2 = \frac{R_BR_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_CR_A}{R_A + R_B + R_C}$$

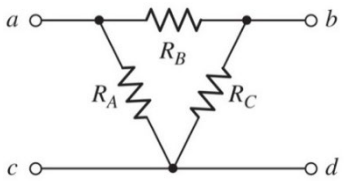
Example: Δ-Y Conversion[1]

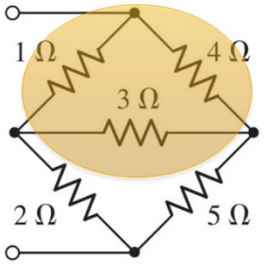
How do we find the equivalent resistance of the following network? Convert a Δ to a Y



Example: Δ-Y Conversion 2

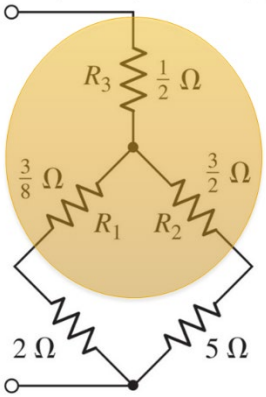
How do we find the equivalent resistance of the following network? Convert a Δ to a Y

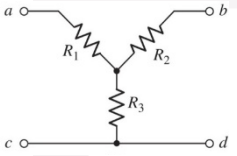




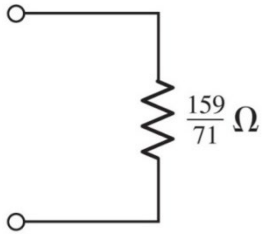
$R_A = 1\Omega$
$R_B = 3\Omega$
$R_C = 4\Omega$

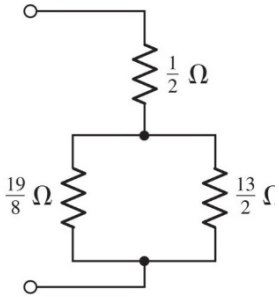
Use the Δ to Y equations





$$R_1 = \frac{R_A R_B}{R_A + R_B + R_C}$$
$$R_2 = \frac{R_B R_C}{R_A + R_B + R_C}$$
$$R_3 = \frac{R_C R_A}{R_A + R_B + R_C}$$

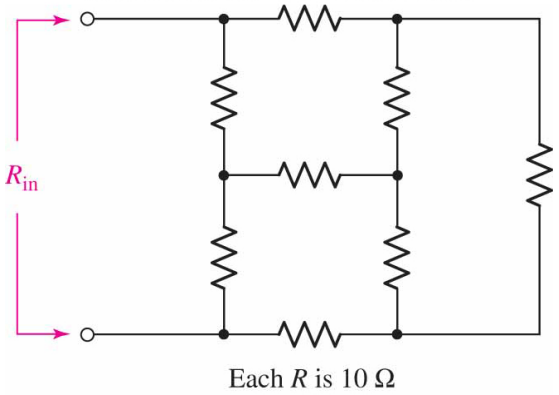


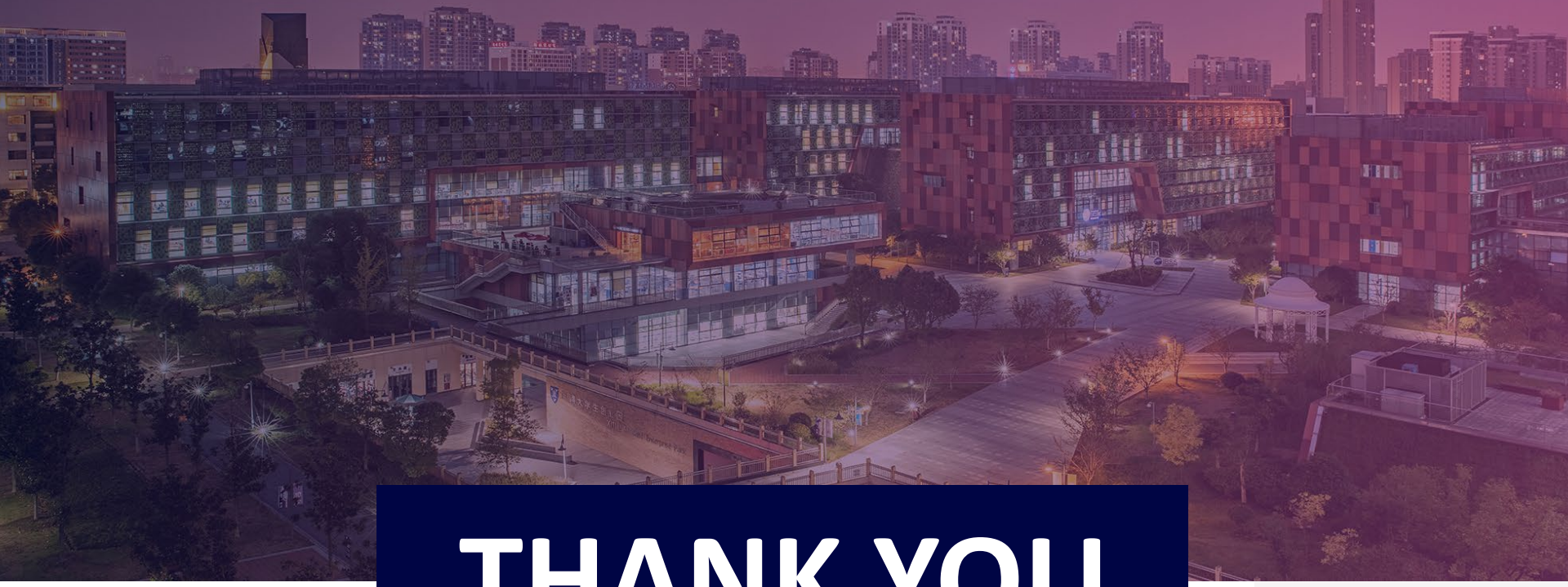


Use standard serial and parallel combinations

Example: Δ-Y Conversion 3

Use the technique of Y–Δ conversion to find the Thévenin equivalent resistance of the circuit





THANK YOU

Xi'an Jiaotong-Liverpool University
西交利物浦大學